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NET512 WEEK 2 PRACTICALS

FIG1: Topology diagram before configuration

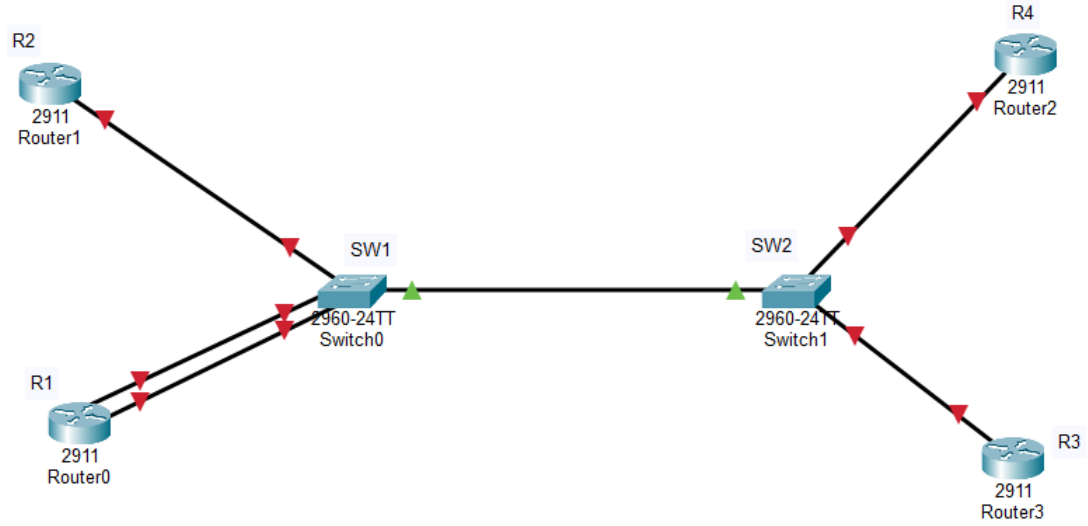


FIG2 :SW1 and SW2 Vlan configurations:

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW1
SW1(config)#vlan 10
SW1(config-vlan)#name R1 - R2
                     ^
% Invalid input detected at '^' marker.

SW1(config-vlan)#name R1-R2
SW1(config-vlan)#vlan 20
SW1(config-vlan)#name R1-R3-R4
SW1(config-vlan)#exit
SW1(config)#int fa0/1
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 10
SW1(config-if)#int fa0/2
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 10
SW1(config-if)#int fa0/3
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 20
SW1(config-if)#int fa0/24
SW1(config-if)#switchport mode trunk

SW1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state
to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state
to up

SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
SW1#
```

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostanme SW2
                ^
% Invalid input detected at '^' marker.

Switch(config)#hostname SW2
SW2(config)#vlan 20
SW2(config-vlan)#name R1-R3-R4
SW2(config-vlan)#exit
SW2(config)#int fa0/3
SW2(config-if)#switchport mode access
SW2(config-if)#switchport access vlan 20
SW2(config-if)#int fa0/4
SW2(config-if)#switchport mode access
SW2(config-if)#switchport access vlan 20
SW2(config-if)#int fa0/24
SW2(config-if)#switchport fa0/24
                          ^
% Invalid input detected at '^' marker.

SW2(config-if)#switchport mode trunk
SW2(config-if)#end
SW2#
%SYS-5-CONFIG_I: Configured from console by console

SW2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
SW2#
```

FIG3: R1,R2,R3 and R4, Task 06 IP address configuration Brief

```
R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.12.1       YES manual up                    up
GigabitEthernet0/1     10.0.13.1       YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

```
R2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R2#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.12.2       YES manual up                    up
GigabitEthernet0/1     10.0.13.3       YES manual down                  down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R2#
```

```
R3#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R3#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  administratively down down
GigabitEthernet0/1     10.0.13.3       YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R3#
```

```
R4#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R4#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.13.4       YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R4#
```

FIG4: R1,R2,R3, and R4 Task 7 routing tables

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter ar
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.12.0/24 is directly connected, GigabitEthernet0/0
L       10.0.12.1/32 is directly connected, GigabitEthernet0/0
C       10.0.13.0/24 is directly connected, GigabitEthernet0/1
L       10.0.13.1/32 is directly connected, GigabitEthernet0/1

R1#
```

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.12.0/24 is directly connected, GigabitEthernet0/0
L       10.0.12.2/32 is directly connected, GigabitEthernet0/0

R2#
```

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter a
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.13.0/24 is directly connected, GigabitEthernet0/1
L       10.0.13.3/32 is directly connected, GigabitEthernet0/1

R3#
```

```
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.13.0/24 is directly connected, GigabitEthernet0/0
L       10.0.13.4/32 is directly connected, GigabitEthernet0/0

R4#
```

FIG5:R2.R3 AND R4 STATIC IP ROUTE CONFIG TASK 08

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip route 10.0.12.0 255.255.255.0 10.10.13.1
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
```

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ip route 10.0.12.0 255.255.255.0 10.0.13.1
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy | Paste

```
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#ip route 10.0.12.0 255.255.255.0 10.0.13.1
R4(config)#end
R4#
%SYS-5-CONFIG_I: Configured from console by console
```

FIG6: Task 9

```
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
S       10.0.12.0/24 [1/0] via 10.0.13.1
C       10.0.13.0/24 is directly connected, GigabitEthernet0/0
L       10.0.13.4/32 is directly connected, GigabitEthernet0/0
```

FIG7: TASK 10 MAC ADDRESS VERIFICATION

```
SW1>en
SW1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0060.702b.3918    DYNAMIC     Fa0/24
  10    0003.e407.1e01    DYNAMIC     Fa0/1
  10    0007.ec62.3501    DYNAMIC     Fa0/2
  20    0003.e407.1e02    DYNAMIC     Fa0/3
SW1#
```

```
SW2>en
SW2#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.96d9.4818    DYNAMIC     Fa0/24
  20    0001.96d9.4818    DYNAMIC     Fa0/24
  20    0090.2b1a.5001    DYNAMIC     Fa0/4
  20    00d0.ba4c.de02    DYNAMIC     Fa0/3
SW2#
```

FIG8:TASK 12 PING VERIFICATION CONNETIVITY

```
R2#ping 10.0.12.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.12.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/6/18 ms
```

```
R3#ping 10.0.13.4

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.13.4, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```

FIG9:TASK 13 DYNAMIC MAC ADDRESS TABLE

```
SW1#show mac address-table dynamic
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0060.702b.3918    DYNAMIC      Fa0/24
  10    0003.e407.1e01    DYNAMIC      Fa0/1
  10    0007.ec62.3501    DYNAMIC      Fa0/2
  20    0003.e407.1e02    DYNAMIC      Fa0/3
  20    0090.2b1a.5001    DYNAMIC      Fa0/24
  20    00d0.ba4c.de02    DYNAMIC      Fa0/24
SW1#
```

```
SW2#show mac address-table dynamic
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.96d9.4818    DYNAMIC      Fa0/24
  20    0001.96d9.4818    DYNAMIC      Fa0/24
  20    0003.e407.1e02    DYNAMIC      Fa0/24
  20    0090.2b1a.5001    DYNAMIC      Fa0/4
  20    00d0.ba4c.de02    DYNAMIC      Fa0/3
SW2#
```

FIG10: TASK 14 MAC ADDRESS CLEARANCE AND VERIFICATION

```
SW2#clear mac address-table dynamic
SW2#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.96d9.4818    DYNAMIC     Fa0/24
  20    0001.96d9.4818    DYNAMIC     Fa0/24
SW2#
```

```
SW1#clear mac address-table dynamic
SW1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0060.702b.3918    DYNAMIC     Fa0/24
  10    0007.ec62.3501    DYNAMIC     Fa0/2
SW1#
```

TASK 15:

After I cleared the MAC address table, the learned entries were gone. They did not come back straight away. A switch only learns a MAC address when a frame comes in from a device, because it reads the source MAC address on that frame. With no traffic, the switch has nothing to learn from.

When the devices send traffic again, for example a ping, ARP, or CDP messages from the routers, the switch learns the MAC addresses again. The entries then show up on the correct ports. This shows that the MAC address table is dynamic. The switch builds it by itself from the traffic it sees, and nobody has to type it in